Blades of Glory? The Impact of Wind Farms on Crime

Jaehyun Choi, Alberto Ortega, and Nikolaos Zirogiannis*

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Abstract

Wind energy has expanded rapidly over the past two decades and accounts for a substantial share of new electricity generation capacity in the United States. A growing literature has documented both the economic benefits of wind energy—such as increased income, GDP, and home values—and its potential health costs, including concerns that low-frequency noise from wind turbines may lead to sleep disruption and increased suicide rates. This paper contributes to this literature by providing the first evidence on the impact of commercial wind turbine installations on local crime rates. Using Uniform Crime Reporting (UCR) data from 1995 to 2021 and a staggered difference-in-differences framework, we estimate the causal effects of wind energy development on a range of criminal offenses. We use the U.S. Census Bureau's Government Finance Database to explore the mechanism through which wind farms affect local revenues and expenditures. We also explore other potential mechanisms, including changes in local population dynamics and migration patterns following wind turbine installation. Our findings indicate that county-level crime rates decrease about 3 years after the installation of wind turbines and local revenue and expenditures increase post-siting. These results highlight important heterogeneity in the social externalities of wind energy and provide valuable considerations for policymakers involved in the siting of renewable energy infrastructure.

Keywords: wind energy, crime, local government revenue

*Choi: O'Neill School of Public and Environmental Affairs, Indiana University - Bloomington, Ohio State University and NBER jc234@iu.edu; Ortega: O'Neill School of Public and Environmental Affairs, Indiana University - Bloomington and NBER, Email: alorte@iu.edu, and Zirogiannis: O'Neill School of Public and Environmental Affairs, Indiana University - Bloomington, Email: nzirogia@iu.edu. The authors declare no known conflicts of interest, financial or otherwise, related to this work.

1 Introduction

Wind energy has experienced significant growth over the past two decades, accounting for a substantial portion of the US energy mix. Since 2000, wind energy has grown at an average annual rate of more than 20%, expanding from just 6 billion kWh in 2000 to over 380 billion kWh in 2021, making it one of the fastest-growing electricity sources in the US (U.S. Energy Information Administration, 2022). This rapid expansion has brought tens of thousands of wind turbines to American landscapes. As of 2024, over 75,000 turbines were operating across 45 states (U.S. Geological Survey et al., 2023). The majority of these installations are in rural counties, where wind farms are often touted as engines of economic development (through job creation, land lease payments, and tax revenues) for communities that have struggled economically in recent decades (**?**). Given this large and growing footprint of wind energy infrastructure, it is increasingly important to understand the broader social impacts of wind farm adoption on local communities. In this paper, we examine one such impact that has received no attention to date: the effect of wind energy development on local crime rates.

Crime is a critical social indicator and a persistent policy concern. Notably, US crime rates have declined substantially over the last 30 years, with both violent and property crime roughly half as prevalent today as in the early 1990s (Pew Research Center, 2020). Despite this long-run decline, public concern about crime remains high. More than three-quarters of Americans – 77% in a recent Gallup survey – believe crime is rising nationally (Jones, 2023). Each year still sees a large volume of offenses: for example, in 2019, there were about 8.1 million reported violent and property crimes in the United States (Federal Bureau of Investigation, 2020). The societal costs of these crimes are substantial, with annual cost of crime estimates at around $2.6 trillion (2017 dollars) when accounting for both direct expenditures and the intangible losses suffered by victims (Miller et al., 2021). Given the scale of crime's impact, even small changes in crime rates can have considerable social welfare implications.

Economic theory offers ambiguous predictions for how a local wind farm development might affect crime. On the one hand, the classic Becker (1968) model of crime as a rational choice implies that improving legitimate economic opportunities should reduce criminal behavior. Wind farm projects often bring an influx of investment and jobs – especially during the construction phase –

which can raise local employment and income (Brunner and Schwegman, 2022; U.S. Department of Energy, 2022; Slattery et al., 2011). Higher incomes and improved job prospects increase the opportunity cost of crime, leading individuals to substitute away from illegal activities in favor of legal work. Conversely, numerous studies have found that local economic booms can coincide with an increase in crime. For example, the opening of casinos has been linked to higher crime rates in host counties (Grinols and Mustard, 2006), and sudden resource booms from oil and gas development have been associated with crime surges (Street, 2025; James and Smith, 2017; Freedman and Owens, 2016). Several mechanisms can explain why crime might increase during booms even if overall economic welfare improves. First, rapid growth can attract an influx of outside workers and new residents, straining local social networks and law enforcement; an inflow of predominantly young male workers (a demographic group with higher crime propensity) can mechanically raise crime rates. Second, greater economic activity creates more targets and opportunities for crime – for instance, more wealth and goods in the community may invite theft. Third, uneven distribution of the boom's benefits can increase income inequality, which in turn may fuel relative deprivation and criminal behavior (Fajnzylber et al., 2002). Additionally, residents with newfound disposable income might engage in the consumption of alcohol or other substances that facilitate crime (Freedman and Owens, 2016).

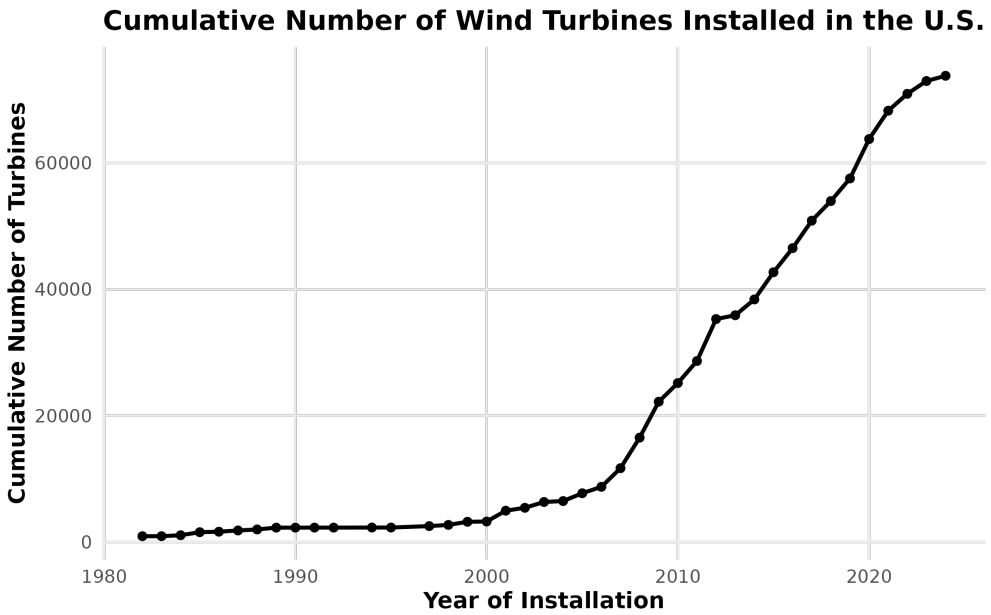
Wind farm developments share many features of local booms – but also some crucial differences – that make their net impact on crime an open empirical question. Like other economic booms, a wind project can inject substantial short-run spending into a community. During the construction period, typically lasting a year or more, large crews of specialized workers may temporarily reside in the area. This temporary shock could lead to some of the "boomtown" effects, including a transient population increase, more frequent interactions among strangers, and a heavier strain on local amenities and law enforcement. Indeed, prior research on energy boomtowns (e.g., shale drilling) has documented spikes in certain crimes (especially disorderly conduct, assaults, and property crimes) attributable to the arrival of out-of-town workers (James and Smith, 2017; Komarek, 2018). However, the wind energy boom differs from oil and gas in that long-term operations are far less labor-intensive – once a wind farm is operational, it provides only a handful of permanent jobs. Thus, any population influx is smaller and shorter-lived compared to fossil fuel booms, potentially limiting the long-term external crime associated with migration. Additionally, the eco-

nomic benefits from wind farms can be widely shared among local landowners through land lease payments, as well as local governments via property tax revenues or payment-in-lieu-of-tax agreements. These steady revenue streams can raise property values and strengthen the local tax base, enabling increased public expenditures on schools, infrastructure, and law enforcement. Consistent with this, recent evidence suggests that wind installations lead to higher incomes and housing values in rural counties on average (**?**Brunner and Schwegman, 2022), indicating improved economic conditions that could potentially favor crime reduction. On net, it is an empirical question whether the pro-social effects of economic improvement outweigh any crime-promoting influences of a wind farm boom. Another factor to consider is the potential for environmental or health externalities from wind turbines that could indirectly affect crime. Wind turbines generate noise (especially low-frequency "infrasound") and visual impacts that have sometimes been associated with stress, annoyance, and health complaints among nearby residents (Zerrahn, 2017). Suppose turbine noise or shadow flicker disrupts sleep or quality of life. In that case, it may lead to behavioral changes – such as increased alcohol use, aggression, or mental distress – that might influence crime. For instance, Zou (2020) finds that wind turbine installations led to higher suicide rates in affected US counties, with the pattern of results pointing toward sleep deprivation from low-frequency noise as a contributing mechanism. While the broader empirical evidence on wind turbines' health impacts is mixed—with most earlier studies reporting no measurable long-term effects on physical or mental health for residents near wind farms (Knopper and Ollson, 2011)—more recent reviews suggest elevated rates of sleep disturbance near turbines and emphasize ongoing gaps in understanding non-acoustic drivers of well-being (Godono et al., 2023). These findings suggest that wind farms may impact community well-being through channels beyond economic development. We employ a difference-in-differences strategy leveraging staggered wind farm adoption across US counties. We leverage data from the FBI's Uniform Crime Reports to examine both violent and non-violent crimes. Thus far, we have found that the presence of wind farms is associated with a long-run decrease in crime.

2 Background

Figure 1 shows the cumulative number of operational wind turbines in the United States from 1982 to 2024. Figure 2 presents the distribution of wind turbine installations from 1982 to 2024.

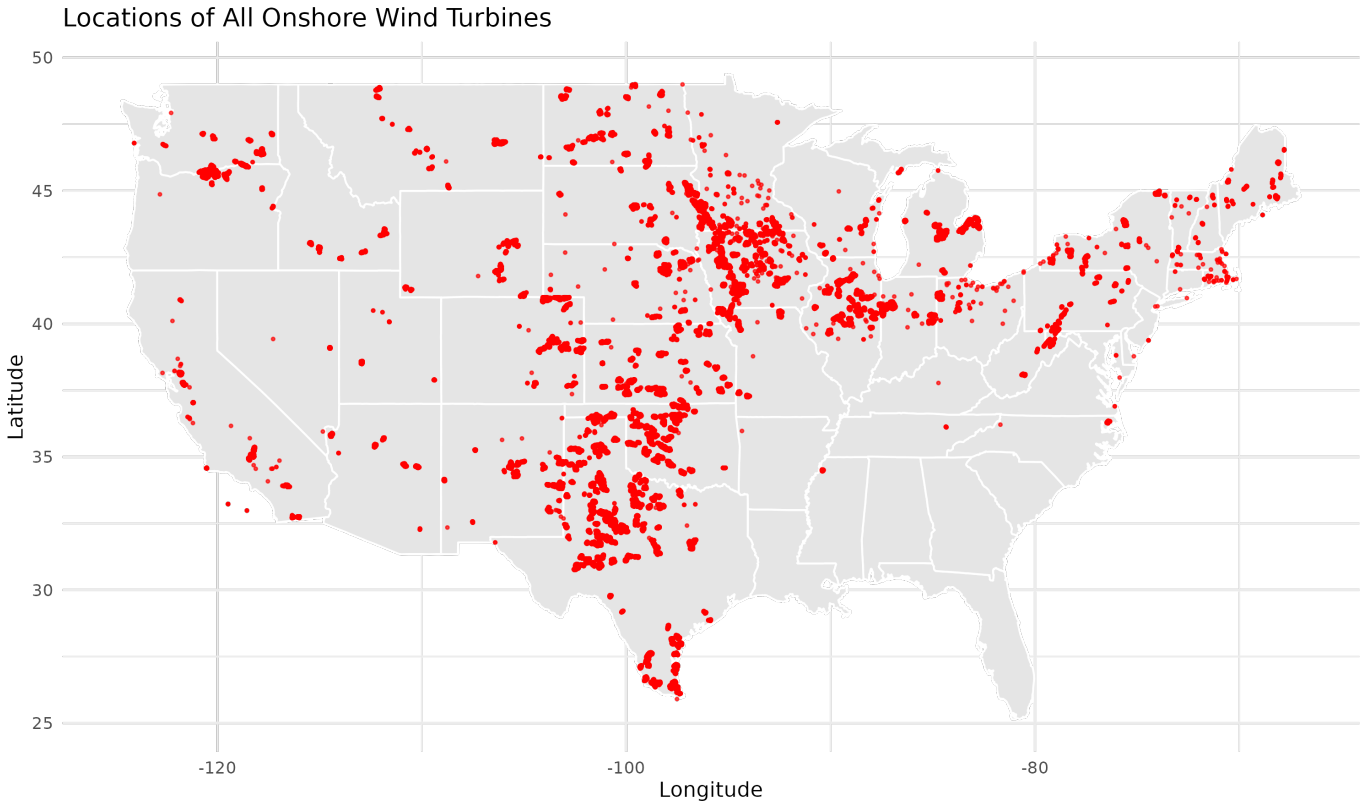
Figure 1: Cumulative Number of Wind Turbines Installed in the United States, 1982–2024.



Interest in wind energy first surged in the 1970s as oil prices spiked, prompting increased federal research and investment in alternative energy technologies. The Public Utility Regulatory Policies Act of 1978 (PURPA) marked a key policy milestone by requiring utilities to purchase electricity from renewable sources, including wind. In 1980, the first large-scale wind farms were installed in California, providing valuable lessons about environmental impacts and turbine siting that shaped later technological improvements. But until 1990s, wind energy production was extremely rare, concentrated almost entirely in California and to a lesser degree in Texas. There were only 7 counties in the US with wind energy installations, producing less than 2 GW of power. Momentum accelerated after the Energy Policy Act of 1992, which introduced a federal production tax credit of 1.5 cents per kilowatt-hour for wind-generated electricity, catalyzing private investment and utility-scale adoption. Growth intensified after 2005, supported by state renewable portfolio standards, declining turbine costs, and continuing federal incentives. The U.S. Department of Energy's 2008 report, 20% Wind Energy by 2030, further signaled national commitment to large-scale wind development. By the 2010s, the United States had more than 60,000 operational turbines, and by 2019 wind energy provided enough electricity to power over 28 million homes, underscoring its emergence as a major component of the nation's energy portfolio. By 2019, installed wind energy generating capacity had grown to over 122 GW and wind energy production had spread across 670

counties and 38 states. The vast majority of these wind energy installations are concentrated in the western, south western, and central United States, and there is significant variation in installed wind energy capacity both across states, as well as across counties within a state (U.S. Department of Energy, 2026).

Figure 2: Wind Turbines Installed in the United States, 1982–2024.

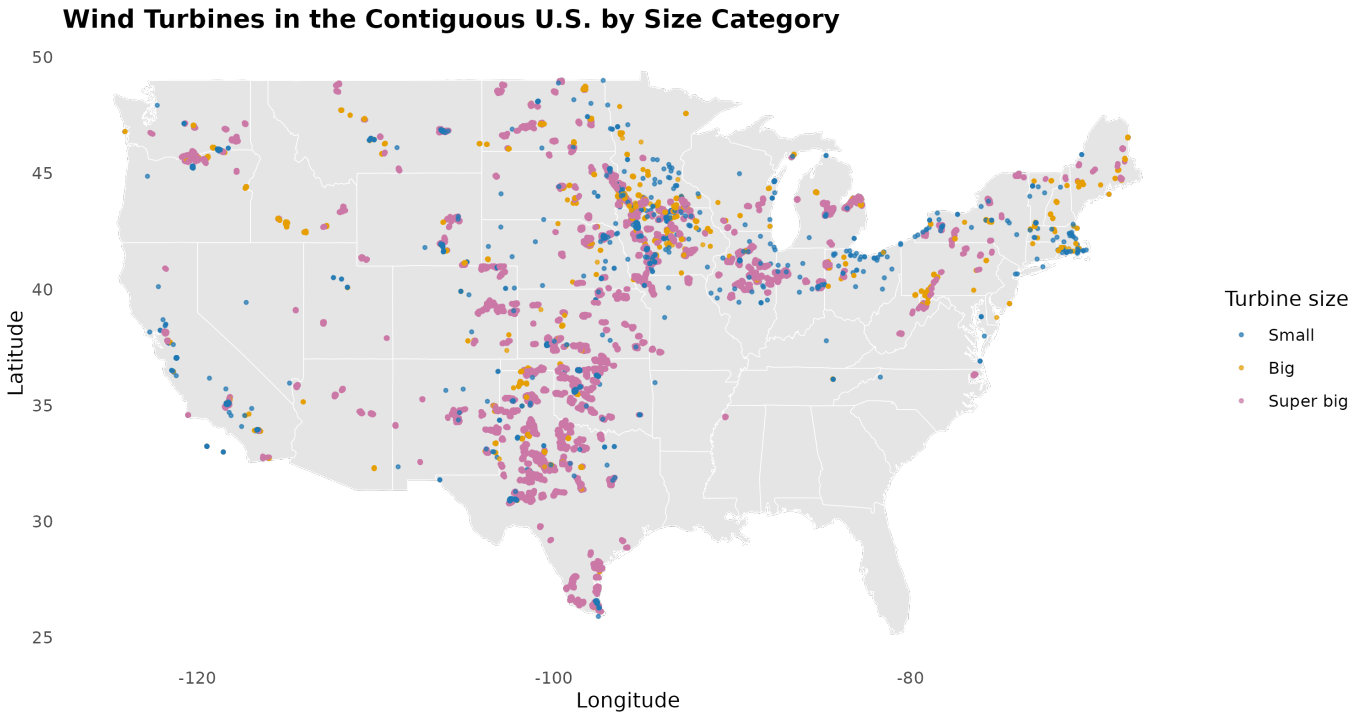


3 Data and Methods

3.1 Wind Farm Installations

We obtain information on wind capacity at the county level from the United States Wind Turbine Database (USWTDB). The USWTDB has information on the precise geolocation of each wind turbine, the year it came into operation as well as it's nameplate capacity in kilowatts. We aggregate the number of operational turbines within each county and year to create a panel dataset showing the cumulative count of wind turbines over time. We then define a treatment indicator that equals one in the first year that a county hosts at least one operational wind turbine and

Figure 3: Wind Turbines Installed in the United States by size, 1982–2024.



remains one thereafter.

3.2 Crime Data

We merge the wind turbine dataset with county-level crime data from the Federal Bureau of Investigation's Uniform Crime Reports (UCR), which provide annual measures of reported offenses by crime category at the police agency level. Reporting frequency varies across agencies—some submit monthly data, while others report annually. To ensure robustness, we construct two versions of the dataset: one using agencies that report in every month of a given year, and another restricted to agencies that report at least once per year. The results are consistent across both specifications.

3.3 Government Finance Data

We incorporate county-level fiscal data from the U.S. Census Bureau's *Annual Survey of State and Local Government Finances* and the *Census of Governments*. These data provide comprehensive information on local government revenues, expenditures, capital outlays, debt, and intergovernmental transfers. From this source, we extract measures such as total revenue, total expenditures,

total capital outlays, and own-source revenues for each county-year.

3.4 Other

For demographic and economic controls, we add county-level population data from the Surveillance, Epidemiology, and End Results (SEER) program, rural–urban continuum codes from the U.S. Department of Agriculture (USDA) to capture variation in urbanization, and county-level gross domestic product (GDP) data from the Bureau of Economic Analysis (BEA).

3.5 Empirical Strategy

Our main empirical approach uses an event study setup, that assesses whether our identifying assumption regarding the effect of wind farm installations on crime holds. We examine how crime rates in county c in state s in year t are affected by wind farm installations that occur in event time h. Since adoption of wind turbines is staggered across time, our event study is estimated using the Sun and Abraham (2021) difference-in-differences estimator. Control counties are those that never experience a wind farm installation in our sample period (1995-2021). Negative values of h indicate years before the treatment. Our baseline for comparison is the year before the first installation occurred (i.e. $h = -1$). The coefficient of interest θ_h, captures the time varying effect of the treatment. If our identifying assumption holds, the coefficients of θ (and associated confidence intervals) for negative values of h should be centered around zero, since there should be no effect from a wind turbine on crime, before it is installed. Coefficients of θ for positive value of h show the effect of the treatment across time. Equation (1) controls for state by year fixed effects ($\gamma_s \times t$) and police agency fixed effects (α_i). Standards errors are clustered at the county level.

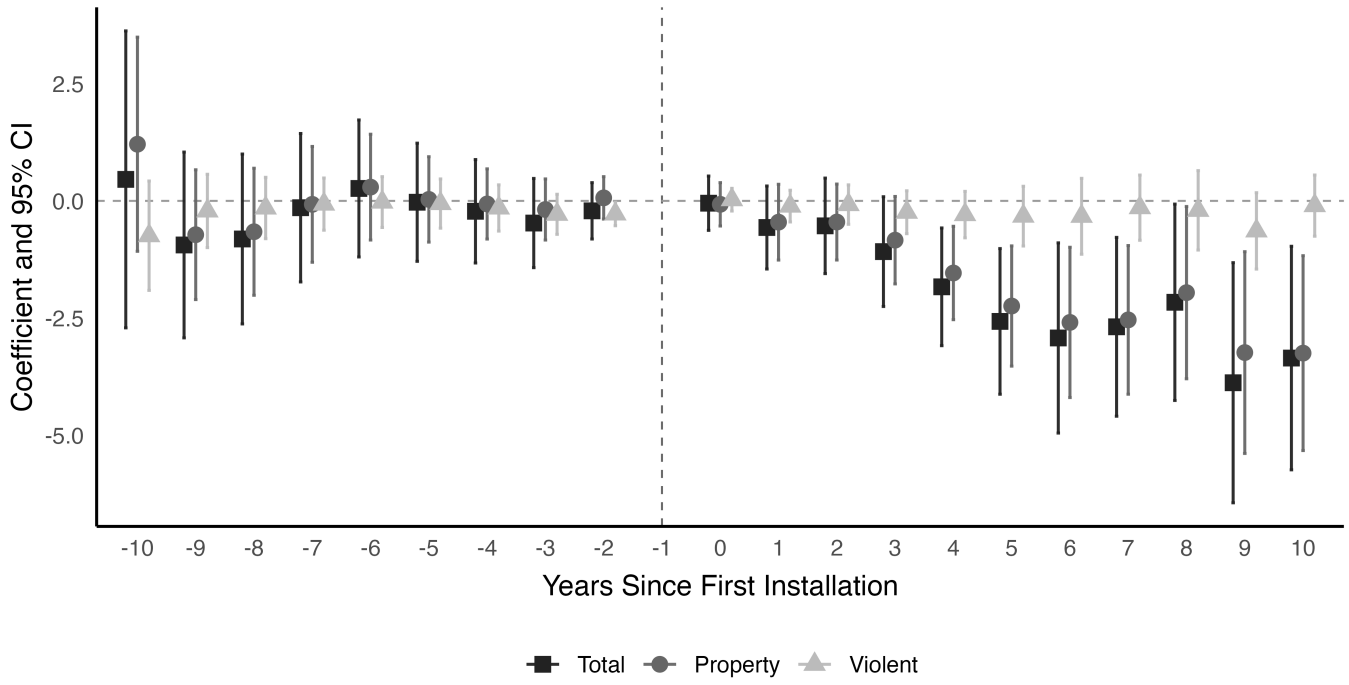
$$Crime_{cst} = \sum_{h=-\kappa_{pre};h\neq-1}^{\kappa_{post}} [\theta_h(WindTurbine_{cs} \times 1[e = h])] + \alpha_i + \gamma_s \times t + m_{ict} \tag{1}$$

4 Results

Figure 4 plots event-study estimates of the effect of wind turbine installation on county-level crime rates, separately for total, property, and violent crimes. The figure uses crimes data for agencies that report every month in the FBI Uniform Crime Report (UCR). The horizontal axis denotes years relative to the first installation, and the vertical axis shows the estimated coefficients.

Coefficients are expressed in crimes per 1,000 residents. The coefficients for the pre-installation period are close to zero and statistically insignificant, indicating no detectable pre-trends in crime rates before the installation of wind turbines. This provides evidence for the validity of our identifying assumption. Beginning roughly three to four years after installation, both total and property crime rates decline steadily. The average treatment effect is a reduction of 2.02 property related crimes and 2.21 total crimes per 1,000 residents. These effects persist for up to a decade after installation. Interestingly, the estimates for violent crime are centered around zero throughout the sample window.

Figure 4: Wind Farms and Total Crime Rates - agencies that report every month to the UCR

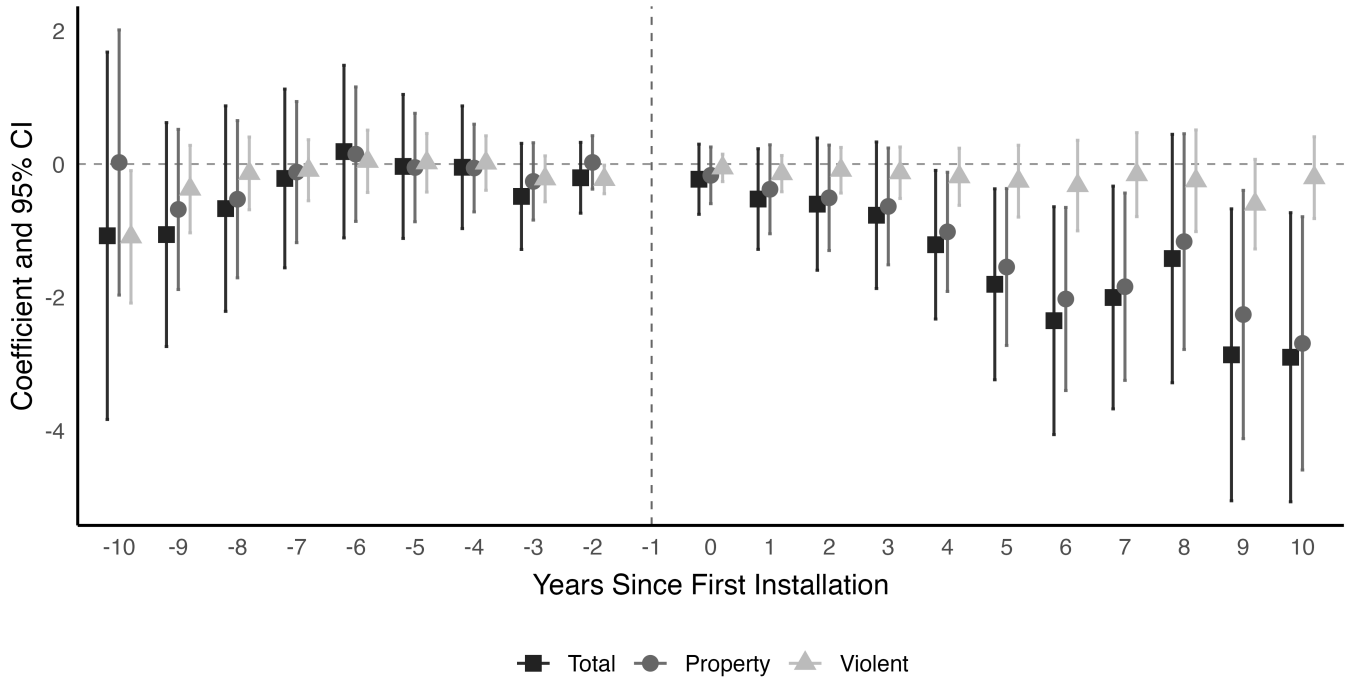


Notes: Using Sun and Abraham (2021) difference-in-differences estimator.

To ensure robustness of these results we perform a separate event study using a group of agencies that report once in every year to the UCR. The results for those agencies, illustrated in Figure 5, are very similar to those in Figure 4. Taken together, our results suggest that wind turbine deployment is associated with a sustained reduction in overall and property-related crime, but has no discernible effect on violent offenses. The absence of pre-treatment trends supports the validity of the identifying assumption, and the delayed yet persistent post-treatment decline is consistent

with longer-run changes in local activity, monitoring, or economic composition following turbine installation.

Figure 5: Wind Farms and Total Crime Rates - agencies that report once in every year to the UCR



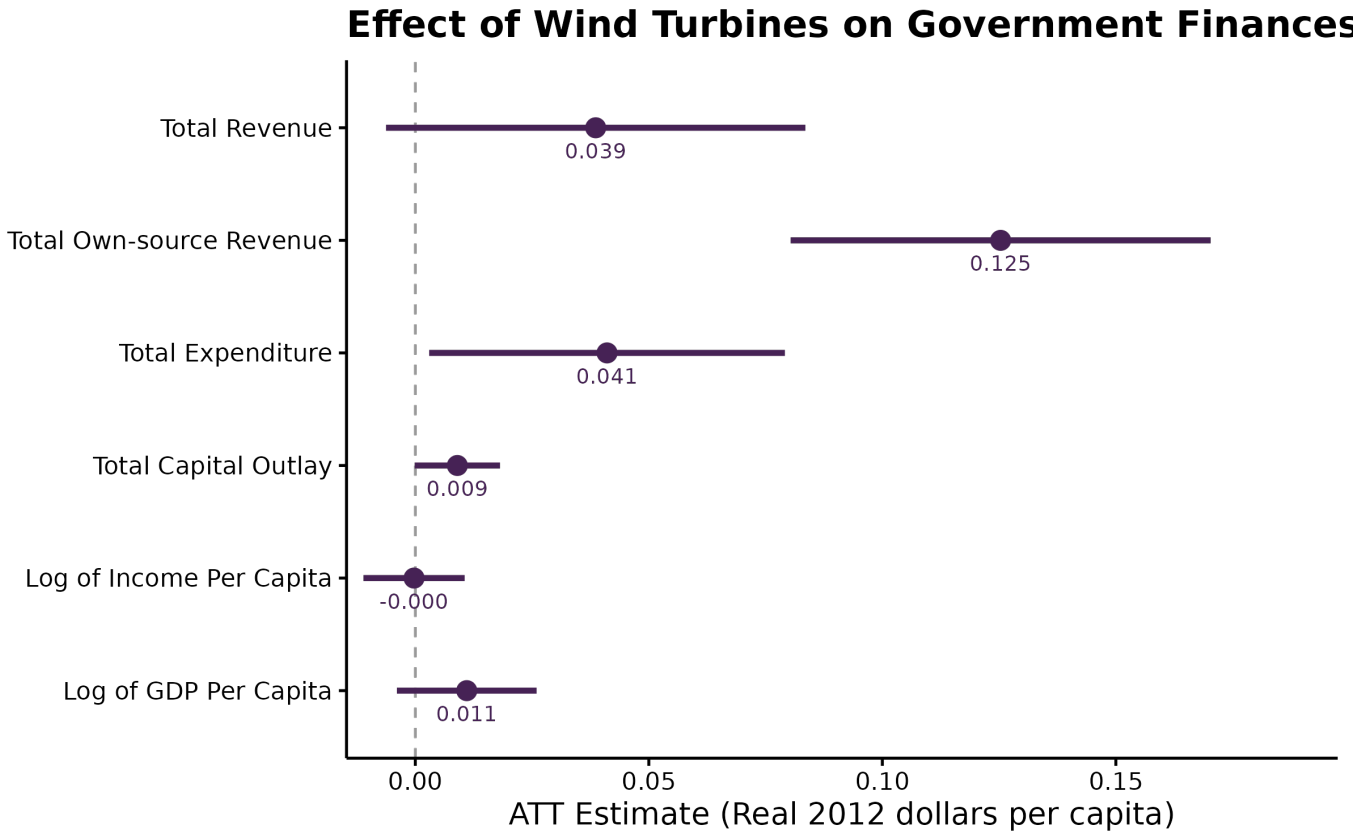
Notes: Using Sun and Abraham (2021) difference-in-differences estimator.

5 Mechanism

An important empirical question is to examine the causal mechanisms that lead to decrease in crime rates. One potential effect is the increased economic activity and revenue from the wind farm installation that serves to increase the opportunity cost of crime. We examine such a mechanism by using data from the U.S. Census Bureau's Government Finance Database. Figure 6 summarizes the average treatment effect results of six difference-in-differences models, that use the same treatment setup as equation (1). We find that wind turbine installations have positive effects across a series of county-wide economic indicators. Specifically, we estimate increases in total revenue, total own-source revenue, total expenditure, total capital outlay and Log of GDP per capita. The difference between revenue and own source revenue is that the later does not include intergovermental grants and funds from state/federal sources. Therefore, own-source revenue exclusively pertains to income

received by the local government. These patterns might suggest that wind turbines expand the local property-tax base and enable counties to invest more in public services and amenities. Such fiscal improvements—along with broader economic gains reflected in higher GDP per capita, likely contribute to the observed reduction in crime.

Figure 6: Effect of Wind Turbines on Government Finances



6 Conclusion - Next steps

Our results suggest that economic expansion, through increased revenue and local expenditures decreased crime rates starting three to four years after the first wind turbine installation in a county. We find effects that persist over time and a clear absence of pre-trends in our staggered adoption event study. One question that we have yet to answer is to extent to which this decline in crime could also be driven by a decrease in population for the treated counties, as documented by (Brunner and Schwegman, 2022). Additional, we will investigate individual subcategories of local expenditures post-treatment such as public welfare, health services, housing and community

development.

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